



MA 255: Number Theory

Course Code: MA 255

Course Name: Number Theory

L-T-P-C: 3-0-0-3

Prerequisites: None

Intended for: B.Tech. in Mathematics and Computing

Mutual Exclusion: MA 552

Distribution: Discipline elective

Approval: 59th BoA (B.Tech Mathematics and Computing elective basket); syllabus approved in 61st BoA

Preamble

Elementary Number Theory is a fundamental area of mathematics that develops deep insight into the structure and properties of integers. Concepts originating from number theory form the theoretical backbone of several areas in modern mathematics and computing, including algorithms, cryptography, coding theory, and computational complexity. The proposed course on Elementary Number Theory is designed to introduce students to core topics such as divisibility, prime numbers, greatest common divisors, congruences, modular arithmetic, Diophantine equations, and arithmetic functions. Emphasis is placed on rigorous reasoning, proof techniques, and the development of algorithmic methods for solving number-theoretic problems.

Course Modules with Quantitative Lecture Hours

- **Unit 1:** Divisibility, The greatest common divisor, The fundamental theorem of integers, Basic properties of congruences, Some special congruences, Polynomial congruences and their solutions, Primitive roots (16 Hours)
- **Unit 2:** Quadratic residues and their basic properties, The Gauss Lemma, The law of quadratic reciprocity, Applications to Diophantine equations, Multiplicative arithmetic functions, The Mobius inversion formula, Perfect and Amicable numbers. (16 Hours)
- **Unit 3:** Integer representation and algorithms, Hashing functions, Pseudorandom numbers, Check digits, Introduction to classic cryptography. (10 Hours)

Text Books

1. William W. Adams, Larry J. Goldstein, *Introduction to Number Theory*, Princeton-Hall Inc., 1976
2. Kenneth H. Rosen, *Discrete Mathematics and its applications*, 8th edition, 2021.